Mohammad Tajwar Chowdhury

AI Engineer

Email | Github (@tajwarchy) | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

AI Engineer specialising in retrieval-augmented generation (RAG), autonomous LLM agents, parameter-efficient fine-tuning (LoRA/PEFT), and multi-agent orchestration. Builds production-grade systems from first principles with a focus on system design, empirical benchmarking, and full observability. Proficient with LangChain, LangGraph, FastAPI, Docker, and the broader MLOps stack.

TECHNICAL SKILLS

AI / LLM Engineering	Retrieval-Augmented Generation (RAG), LLM Fine-Tuning, LoRA, PEFT, Autonomous Agents, ReAct Pattern, Multi-Agent Systems, LangChain, LangGraph, MCP, Prompt Engineering
Vector & Memory	FAISS, Qdrant, ChromaDB, Sentence-Transformers, Semantic Search, Embedding Pipelines, HyDE, MMR Retrieval
MLOps & Serving	FastAPI, Docker, Docker Compose, MLflow, Celery, Redis, Prometheus, Grafana, SSE Streaming, Blue-Green Deployment, RAGAS Evaluation
Models & Frameworks	HuggingFace Transformers, PyTorch, Ollama, Mistral 7B, TinyLlama, OpenAI API, Anthropic API
Computer Vision	YOLOv8, Multi-Object Tracking, StrongSORT, OSNet Re-ID, OpenCV, TrackEval
Core	Python, SQL, REST API Design, System Design, Git

PROJECTS

RAG from Scratch — FAISS · Sentence-Transformers · Mistral 7B — GitHub

No-framework RAG pipeline built entirely from scratch to expose every internal component.

- Built chunking (fixed/sentence/sliding), FAISS indexing, Sentence-Transformers embeddings, and Ollama-based generation without framework abstractions.
- Achieved **93.3% Recall@5** and 0.628 MRR on a benchmark dataset; sentence-aware chunking outperformed alternative strategies.
- Implemented embedding caching with a **60% hit rate**, reducing repeated-query retrieval latency from $\sim$4 s to $<$16 ms.
- Designed a provider-agnostic LLM layer enabling seamless migration between Ollama and OpenAI.

Autonomous AI Agent — ReAct Pattern — GitHub

Production-grade ReAct agent with semantic memory, fault tolerance, and observability.

- Engineered a think–act–observe loop with circuit breakers and loop-detection safeguards.
- Built a dual-storage architecture using ChromaDB for semantic memory and SQLite for audit logging.
- Integrated Prometheus and Grafana monitoring for full visibility into reasoning traces and tool usage.

Production RAG — LangChain · Qdrant · Celery — GitHub

Scalable multi-user retrieval system with asynchronous ingestion and evaluation pipelines.

- Developed a stateless FastAPI service with Celery workers, Redis queues, and Docker Compose deployment.
- Implemented Similarity, MMR, and HyDE retrieval strategies with cross-encoder reranking.
- Evaluated retrieval quality end-to-end using RAGAS metrics.

LLM Fine-Tuning with LoRA — TinyLlama — GitHub

Parameter-efficient LLM fine-tuning workflow with model lifecycle management.

- Fine-tuned TinyLlama using LoRA/PEFT, reducing perplexity from 3.00 to 1.88 (**37% improvement**).
- Managed model versioning with MLflow and deployed inference services via streaming FastAPI.
- Documented blue-green deployment and production-serving considerations.

Multi-Agent System — LangGraph · MCP — GitHub

Supervisor-agent architecture coordinating specialist agents through MCP tooling.

- Built a LangGraph supervisor routing requests among Research, Calculator, and Summarizer agents.
- Used MCP to decouple tool execution from agent logic for improved modularity and testing.
- Added semantic memory with ChromaDB and observability logging through SQLite.

ADDITIONAL PROJECTS

Advanced Multi-Object Tracking YOLOv8m + StrongSORT with OSNet appearance embeddings on MOT17, benchmarked against ByteTrack using TrackEval: **HOTA 41.6 · MOTA 38.1 · IDF1 50.8.** GitHub

Multi-Camera People Tracking Cross-camera re-identification and tracking pipeline across multiple viewpoints using YOLOv8 and Re-ID embeddings. GitHub

EDUCATION

B.Sc. in Computer Science *Class of 2025*

BRAC University, Dhaka, Bangladesh

REFERENCE

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